

**YEAR 11 PHYSICS ATAR**  
**Investigation 4.2 - Latent Heat of Fusion**  
**Marking Rubric**

NAME: \_\_\_\_\_

|                     |                                                                                                                    | Marks<br>Allocated | Mark      |
|---------------------|--------------------------------------------------------------------------------------------------------------------|--------------------|-----------|
| <b>Aim</b>          | Clear aim given.                                                                                                   | <b>1</b>           |           |
| <b>Method</b>       | Refers to laboratory manual for method or gives a suitable method.                                                 | <b>1</b>           |           |
| <b>Results</b>      | Suitable table or similar.                                                                                         | <b>3</b>           |           |
|                     | Correct units                                                                                                      |                    |           |
|                     | Correct significant figures.                                                                                       |                    |           |
| <b>Calculations</b> | Correct setting out.                                                                                               | <b>3</b>           |           |
|                     | Correct calculation.                                                                                               |                    |           |
|                     | Correct units and significant figures.                                                                             |                    |           |
| <b>Discussion</b>   | Q3 Sources of error (at least 2).<br>Estimate of % uncertainty for each source of error.<br>Overall % uncertainty. | <b>3</b>           |           |
|                     | Q4 Impurities - lowers freezing point.<br>Assume melting point = 0°C.<br>Value for $L_f$ is lower.                 | <b>3</b>           |           |
|                     | Q5 Above room temperature - greater $\Delta T$ / negates the effect of heat entering the system from outside.      | <b>1</b>           |           |
|                     | Q6 States accepted value.<br>Calculates % difference.                                                              | <b>2</b>           |           |
|                     | Q7 $E_p$ of particles changes - move further apart.                                                                | <b>1</b>           |           |
|                     | Q8 Ice melts first, then changes temperature.<br>Requires more heat, which is removed from the skin.               | <b>2</b>           |           |
| <b>Conclusion</b>   | States value of latent heat of fusion.                                                                             | <b>1</b>           |           |
|                     |                                                                                                                    | <b>TOTAL</b>       | <u>21</u> |